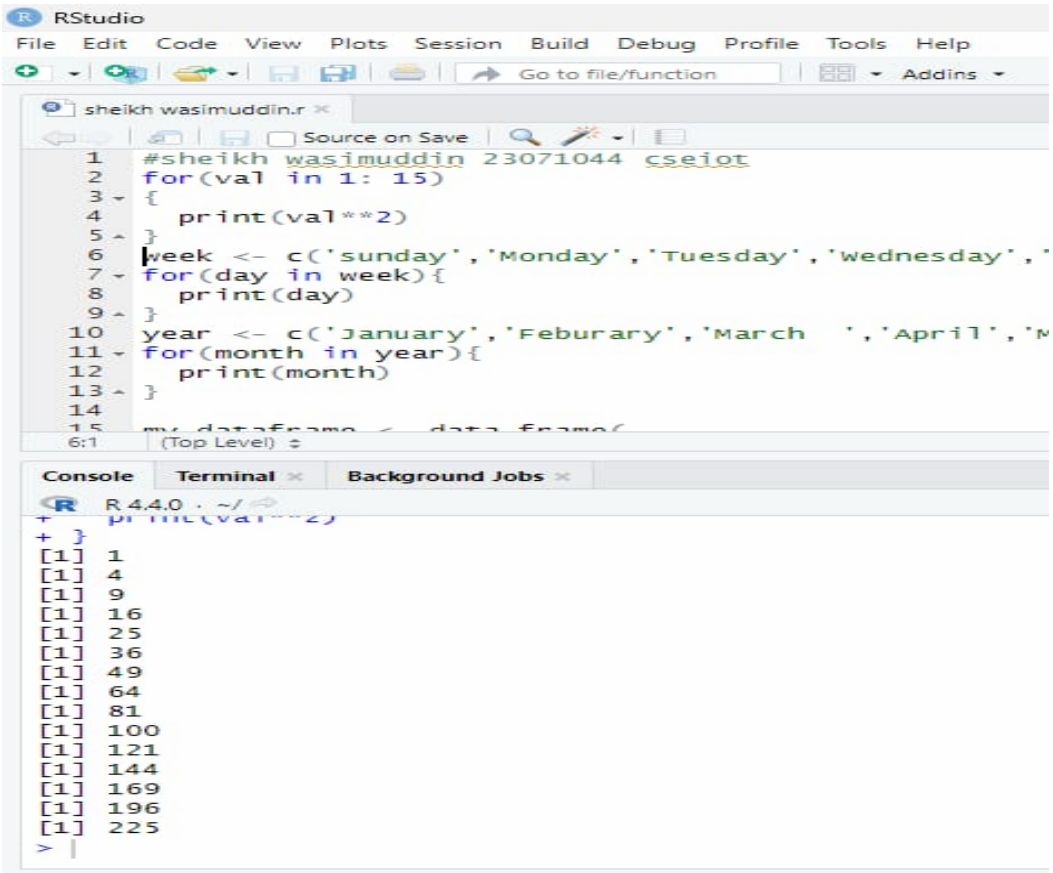
Name: Sheikh Wasimuddin
Roll No:73
Branch:CSE(IoT)

LAB 04(B)	
Aim:	Implementation of loops in R

Theory	In R programming, we require a control structure to run a block of code multiple times. Loops come in the class of the most fundamental and strong programming concepts. A loop is a control statement that allows multiple executions of a statement or a set of statements. The word 'looping' means cycling or iterating.
	
	A loop asks a query, in the loup structure. If the answer to that query requires an action, it will be executed. The same query is asked again and again until further action is taken. Any time the query is asked in the loop, it is known as an iteration of the loop. There are two components of a loop, the control statement, and the loop body. The control statement controls the execution of statements depending on the condition and the loop body consists of the set of statements to be executed.
	
	In order to execute identical lines of code numerous times in a program, a programmer can simply use a loop.
	
	There are three types of loops in R programming:
	
	❖ For Loop
	
	❖ While Loop
	
	❖ Repeat Loop
	
	<u>Syntax:</u>
	
	❖ for (value in sequence)
	{
	statement
	}
	❖ while (condition)
	{
	statement
	}

Name: Sheikh Wasimuddin
Roll No:73
Branch:CSE(IoT)

	❖ repeat
	{
	statement
	if(condition)
	{
	break
	}

Output & Program	❖ FOR LOOP 

Name: Sheikh Wasimuddin
Roll No:73
Branch:CSE(IoT)

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins

sheikh wasimuddin.r

Source on Save | Run | Source

```
week <- c('sunday','Monday','Tuesday','Wednesday','Thursday','Friday','Saturday'
for(day in week){
  print(day)
}
year <- c('January','Feburary','March  ','April','May','June','July',"August","S
for(month in year){
  print(month)
}
```

10:1 (Top Level) R Scr

Console Terminal Background Jobs

R 4.4.0 · ~/

```
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[workspace loaded from ~/.RData]

> week <- c('sunday','Monday','Tuesday','Wednesday','Thursday','Friday','Saturday')
> for(day in week){
+   print(day)
+ }
[1] "sunday"
[1] "Monday"
[1] "Tuesday"
[1] "Wednesday"
[1] "Thursday"
[1] "Friday"
[1] "Saturday"
>
```

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins

sheikh wasimuddin.r

Source on Save | Run | Source

```
for(month in year){
  print(month)
}

my_dataframe <- data.frame(
  Name = c("Joy", "Juliya", "Boby", "Marry"),
  Age= c(50,30,20,10),
  Gender= c("M", "F", "M", "F")
```

15:1 (Top Level) R

Console Terminal Background Jobs

R 4.4.0 · ~/

```
[1] "Tuesday"
[1] "Wednesday"
[1] "Thursday"
[1] "Friday"
[1] "Saturday"
> year <- c('January','Feburary','March  ','April','May','June','July',"August","Sep
ber","October","November","december")
> for(month in year){
+   print(month)
+ }
[1] "January"
[1] "Feburary"
[1] "March  "
[1] "April"
[1] "May"
[1] "June"
[1] "July"
[1] "August"
[1] "September"
[1] "October"
[1] "November"
[1] "december"
>
```

Name: Sheikh Wasimuddin
Roll No:73
Branch:CSE(IoT)

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins

sheikh wasimuddin.r

Source on Save

```
)
print(my_dataframe)
for(i in seq_len(nrow(my_dataframe))){
  current_row <- my_dataframe[i, ]
  print(paste("The current row is:",toString((current_row))))
}
for(j in seq_len(ncol(my_dataframe))) {
  current_column <- my_dataframe[[j]]
  print(paste("The current column is:", toString(current_column)))
}
```

29:1 (Top Level)

Console | Terminal | Background Jobs

R 4.4.0 · ~/

```
6  omkar  18       M
> for(i in seq_len(nrow(my_dataframe))){
+   current_row <- my_dataframe[i, ]
+   print(paste("The current row is:",toString((current_row))))
+ }
[1] "The current row is: sheikh, 19, M"
[1] "The current row is: tanmay, 20, M"
[1] "The current row is: tejas, 21, M"
[1] "The current row is: vedant, 19, M"
[1] "The current row is: parth, 19, M"
[1] "The current row is: omkar, 18, M"
> for(j in seq_len(ncol(my_dataframe))) {
+   current_column <- my_dataframe[[j]]
+   print(paste("The current column is:", toString(current_column)))
+ }
[1] "The current column is: sheikh, tanmay, tejas, vedant, parth, omkar"
[1] "The current column is: 19, 20, 21, 19, 19, 18"
[1] "The current column is: M, M, M, M, M, M"
>
```

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins

sheikh wasimuddin.r

Source on Save | Run

```
year <- c('January','Feburary','March  ','April','May','June','July',"August","September","October","November","december")
for(month in year){
  print(month)
}

my_dataframe <- data.frame(
  Name = c("sheikh", "tanmay", "tejas", "vedant","parth","omkar"),
  Age= c(19,20,21,19,19,18),
  Gender= c("M", "M", "M", "M","M","M")
)
print(my_dataframe)
for(i in seq_len(nrow(my_dataframe))){
  current_row <- my_dataframe[i, ]
  print(paste("The current row is:",toString((current_row))))
}
```

21:1 (Top Level)

Console | Terminal | Background Jobs

R 4.4.0 · ~/

```
> my_dataframe <- data.frame(
+   Name = c("sheikh", "tanmay", "tejas", "vedant","parth","omkar"),
+   Age= c(19,20,21,19,19,18),
+   Gender= c("M", "M", "M", "M","M","M")
+ )
> print(my_dataframe)
    Name Age Gender
1 sheikh  19      M
2 tanmay  20      M
3  tejas  21      M
4 vedant  19      M
5  parth  19      M
6  omkar  18      M
>
```

Name: Sheikh Wasimuddin
Roll No:73
Branch:CSE(IoT)

❖ WHILE LOOP

sheikh wasimuddin.r

Source on Save

```
i <- 1
while (i < 18) {
  print(i)
  i <- i + 1
}
```

6:1 (Top Level)

Console Terminal Background Jobs

R 4.4.0 · ~/

```
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

> i <- 1
> while (i < 18) {
+   print(i)
+   i <- i + 1
+ }
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
[1] 6
[1] 7
[1] 8
[1] 9
[1] 10
[1] 11
[1] 12
[1] 13
[1] 14
[1] 15
[1] 16
[1] 17
>
```

Name: Sheikh Wasimuddin
Roll No:73
Branch:CSE(IoT)

sheikh wasimuddin.r

Source on Save

```r
i <- 18
while (i >0) {
  print(i)
  i <- i - 1
}
```

6:1 (Top Level)

Console Terminal Background Jobs

R 4.4.0 · ~/

```
[1] 9
[1] 10
[1] 11
[1] 12
[1] 13
[1] 14
[1] 15
[1] 16
[1] 17
> i <- 18
> while (i >0) {
+   print(i)
+   i <- i - 1
+ }
[1] 18
[1] 17
[1] 16
[1] 15
[1] 14
[1] 13
[1] 12
[1] 11
[1] 10
[1] 9
[1] 8
[1] 7
[1] 6
[1] 5
[1] 4
[1] 3
[1] 2
[1] 1
>
```

Name: Sheikh Wasimuddin
Roll No:73
Branch:CSE(IoT)

sheikh wasimuddin.r

Source on Save

```
i = readline(prompt = "Enter any number : ");
i = as.integer(i);
print(i)
while (i>0) {
  print(i)
  i = i - 1
}
```

8:1 (Top Level)

Console Terminal Background Jobs

R 4.4.0 · ~/

```
[1] 5
[1] 4
[1] 3
[1] 2
[1] 1
> i = readline(prompt = "Enter any number : ");
Enter any number : 20
> i = as.integer(i);
> print(i)
[1] 20
> while (i>0) {
+   print(i)
+   i = i - 1
+ }
[1] 20
[1] 19
[1] 18
[1] 17
[1] 16
[1] 15
[1] 14
[1] 13
[1] 12
[1] 11
[1] 10
[1] 9
[1] 8
[1] 7
[1] 6
[1] 5
[1] 4
[1] 3
[1] 2
[1] 1
```

Conclusion	Implementation of loops in R successfully executed
Name Of Faculty:	Prof. Akhil Jajulwar